

Label-Efficient Dark-Vessel Detection in SAR: Does SAR-Domain Pretraining Beat Optical, and Does It Hold Across ViT and CNN?

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Abstract

Detecting “dark” vessels—ships not broadcasting their Automatic Identification System (AIS) signal—in spaceborne synthetic aperture radar (SAR) is operationally important and label-scarce. Pretrained backbones can help, but it is unclear which pretraining recipe transfers best, and whether the answer is the same for Transformer and convolutional backbones. We propose a controlled label-efficiency study on xView3-SAR [8] that holds one point-native detector fixed and varies only the downloaded backbone initialization, across two size-matched tracks (ViT-B/16, $\sim 86\text{M}$; ConvNeXt-V2-Base, $\sim 89\text{M}$) and four pretraining recipes each: random, optical, SAR, and supervised out-of-domain SAR. Because every backbone is a released checkpoint, differences are attributable to the pretraining source rather than to our own optimization choices.

1 Introduction

Detecting ships not broadcasting their AIS signature—“dark vessels”—in spaceborne SAR is a critical maritime-monitoring problem with scarce labels: xView3-SAR [8] provides Sentinel-1 imagery, but its human-verified dark-vessel labels are few, so supervised-only training overfits.

Pretrained backbones can help, but it is unclear which pretraining recipe transfers best: self-supervision on generic optical satellite imagery (fMoW-RGB [2] via SatDINO [9]), self-supervision on large-scale SAR (SAR-1M via SARMAE [7]), or supervised detection on labeled out-of-domain SAR (LS-SSDD [11])—and whether the best recipe is the same for a ViT [4] and a ConvNeXt [10]. We compare released checkpoints rather than pretraining our own so that each arm faithfully represents its recipe as published; re-pretraining would confound the pretraining source with our own optimization choices [13].

With the detector, schedule, and input fixed, we ask: (Q1) how much labeled xView3 data does each recipe save relative to random initialization to reach a target dark-vessel F1; (Q2) does SAR-domain pretraining beat

optical in the scarce-label regime; and (Q3) is that ordering architecture-general, holding for both a ViT and a size-matched CNN? We hypothesize that, within each track, SAR-domain pretraining yields the steepest low-label gains (random $<$ optical $\lesssim$ supervised $<$ SAR), and that this ordering holds for both architectures.

2 Approach

Design. The study has two architecture-matched tracks—ViT-B/16 ($\sim 86\text{M}$) and ConvNeXt-V2-Base ($\sim 89\text{M}$, deliberately size-matched)—each with four arms that share the same head, optimizer, schedule, and seeds; only the downloaded initialization differs (Table 1). Initialization is the only variable within a track and architecture the only variable across matched roles, so comparing arm k to arm $k+4$ isolates the architecture family with the pretraining role held fixed.

Pretraining regimes. The recipes differ in modality, labels, and domain: optical (SatDINO, DINO [1]; BigEarthNet-S2) is a different sensing modality (no speckle, different scattering physics); SAR, self-supervised (SARMAE, an MAE [5]; BigEarthNet-S1) is the same modality but pretrained without detection labels on a different scene distribution; and labeled out-of-domain SAR (LS-SSDD) is the same modality and a related supervised task, but a different acquisition. Q2 thus contrasts “same pixels, different task” against “different pixels, same task.”

Detector. xView3 labels are points and its metric matches predictions by distance, not box IoU, so we use a point-native heatmap head in the style of CenterNet / Objects as Points [12]: each vessel is a Gaussian blob on a stride-4 map, decoded by peak-finding with distance-based non-maximum suppression. Every arm receives the same fixed 3-channel input [VH, VV, VH–VV] in dB, so input handling is never a confound between arms.

Arm	Track	Backbone	Initialization	Params	On curve?	Role
1	ViT	ViT-B/16	random init	~86M	yes	ViT floor
2	ViT	ViT-B/16	SatDINO (fMoW-RGB, DINO)	~86M	yes	optical-domain transfer
3	ViT	ViT-B/16	SARMAE (SAR-1M, MAE)	~86M	yes	SAR-domain transfer
4	ViT	ViT-B/16	LS-SSDD supervised (we train)	~86M	yes	labeled out-of-domain SAR
5	CNN	ConvNeXt-V2-B	random init	~89M	yes	CNN floor
6	CNN	ConvNeXt-V2-B	BigEarthNet-S2 (optical RS)	~89M	yes	optical-domain transfer
7	CNN	ConvNeXt-V2-B	BigEarthNet-S1 (SAR RS)	~89M	yes	SAR-domain transfer
8	CNN	ConvNeXt-V2-B	LS-SSDD supervised (we train)	~89M	yes	labeled out-of-domain SAR
R1	ref	ConvNeXt-V2-B	ImageNet (contingent)	~89M	no	generic-transfer reference
R2	ref	YOLO26	COCO-pretrained	external	no	detector reference
R3	ref	LocateAnything-3B	zero-shot	~3B	no	VLM reference

Table 1: Eight controlled arms in two size-matched tracks (~86M vs. ~89M), plus three external references reported off the curves. Optical anchors are SatDINO [9] (ViT) and BigEarthNet-S2 [3] (CNN); SAR anchors are SARMAE [7] (ViT) and BigEarthNet-S1 [3] (CNN).

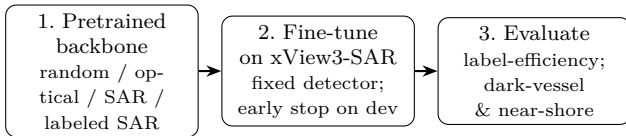


Figure 1: The three stages, run identically for every arm; only stage 1 (the downloaded backbone) varies within a track.

Fine-tuning and comparability. All arms fine-tune end-to-end through one code path with shared AdamW, augmentation, and seeds, and are trained until dev-split F1 stops improving (early stopping), not for a fixed epoch budget; the decision threshold is selected on dev and frozen before test. Continuous-integration guard tests make comparability a checked invariant rather than a claim: a backbone-parity test asserts identical within-track parameter counts and matching head-adapter geometry, a checkpoint-load test asserts each backbone loads its pretrained weights (not silently random), and a scorer-immutability test freezes the evaluation code (Figure 1).

3 Data and Evaluation

We use xView3-SAR [8] via the SARFish mirror (Sentinel-1 GRD, dual-polarization VH/VV): 75–150 training scenes chipped at 800×800 ($\sim 10^4$ chips), sub-sampled at the scene level to 10/25/50/100% of labels and split 75/15/10 (train/dev/test); the ~50 human-verified scenes are a disjoint, once-touched final-evaluation set. Arms 4 and 8 pretrain on LS-SSDD-v1.0 [11] (~9,000 labeled 800×800 sub-images), used only as a pretraining source and never mixed into the xView3 splits. The headline result is a two-track label-efficiency curve (detection F1 vs. 10/25/50/100% labels; eight lines) plus dark-vessel recall and near-shore F1, the operationally hardest slices: the low-label gap over the random floor answers Q1–Q2, and matched-role gaps across

tracks answer Q3. External references (YOLO26 [6], a zero-shot VLM, and a contingent ImageNet-ConvNeXt) situate the eight arms without entering the controlled comparison.

4 Timeline and Feasibility

Two contributors work independent lanes (data/scorer; detector/eval) with no pretraining handoff to block on. Because all foundation-model backbones are downloaded, compute is modest: the full eight-arm grid with seed re-runs plus the two small LS-SSDD supervised backbones is ~293–360 GPU-hours (~3–4 nights on one $8 \times V100$ node)—we download checkpoints for comparability of recipes as published, and the low cost is a side benefit. The downloaded-backbone arms run first to de-risk the fixed-input channel format. We target the WACV 2027 Round-2 deadline (paper registration August 21, 2026; submission August 28, 2026, AoE).

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